

Way-Finder Haptic Insoles

Cost Analysis per Unit — CSMFAB 038

Category: Aegis Embark — Footwear **Unit Basis:** 1 pair + LoRa navigation module **Date:** June 2026 | **Document Type:** CSMFAB Cost Study V2.0

1. Product Summary

This cost analysis is derived from the V2.0 specification and v1to2 versioning document for this product. All material costs reflect 2026 market pricing. Three production scales are costed: - **Prototype:** 1-5 units (single-setup, high labor, no tooling amortization) - **Small Batch:** 100-500 units/year (semi-tooled, moderate labor) - **Production Volume:** 5,000-50,000 units/year (fully tooled, optimized labor)

2. Bill of Materials (BOM)

Component	Quantity/Unit	Unit Price	Extended Cost
PVDF-TrFE harvesting stack (per pair, 24 elements)	0.04 kg	\$350.00/kg	\$14.00
ZrB2-SiC ceramic heel insert (pair)	0.2 kg	\$350.00/kg	\$70.00
KNbO3-BaTiO3 haptic actuator array (8 elements)	0.02 kg	\$180.00/kg	\$3.60
LoRa radio module	0.02 kg	\$18.00/kg	\$18.00
MRAM offline map chip (4GB)	0.005 kg	\$45.00/kg	\$45.00
IMU sensor module (PEEK-housed)	0.01 kg	\$25.00/kg	\$25.00
Standard insole base (dielectric EVA)	0.1 kg	\$8.00/kg	\$0.80
Aerogel heel pad (polyimide-silica)	0.05 kg	\$250.00/kg	\$12.50
Micro-LiFePO4 cell (30mAh)	0.01 kg	\$100.00/kg	\$1.00
BOM Subtotal			\$189.90

3. Labor, Process & Overhead Costs

Cost Element	Prototype	Small Batch	Production Volume
BOM Materials	\$189.90	\$155.72	\$123.44
Direct Labor	\$160.00	\$108.00	\$72.00
Overhead (15% of labor)	\$24.00	\$16.20	\$10.80
Quality Control & Testing	\$19.20	\$8.64	\$3.60
Packaging & Logistics	\$25.00	\$20.00	\$15.00

Tooling & Equipment Notes: Harvester stack assembly: \$35/pair. MRAM chip is standard electronics component.

4. Per-Unit Cost Summary

Scale	Total COGS	Suggested MSRP Range	Gross Margin
Prototype (1-5 units)	\$880	\$1,408 – \$1,936	35-55%
Small Batch (100-500/yr)	\$590	\$885 – \$1,180	33-50%
Production Volume (5K-50K/yr)	\$420	\$588 – \$756	28-45%

Market Reference: \$480-700/pair (smart insole market \$80-500; GPS-free + harvesting = differentiated premium)

5. Key Cost Drivers (V2.0)

Driver	Impact	Mitigation
ZrB2/Si3N4 UHTC ceramics	HIGH — specialty powder \$350+/kg	Bulk procurement; domestic synthesis pilot (Japan NIMS CO2 route)
MXene Ti3C2Tx	VERY HIGH — \$2,000/kg at 2026 scale	Tile pattern minimizes usage; price projected -60% by 2028 as production scales
Elium® thermoplastic BFRP	MEDIUM — recyclability adds value; \$12/kg resin	Phoenix Protocol recycling recovers 100% monomer; net cost offset
Tooling NRE	HIGH for first production run	Paid by pre-orders or grant funding; amortized quickly
Flash Sintering process	MEDIUM — shares SPS equipment	Regional ceramic sintering facility sharing model

6. Phoenix Protocol Circular Economy Value

At end-of-life, material recovery adds back-value: - **Recovered material value per unit:** \$34 – \$66 - **Recovery reduces net lifecycle cost by:** 15-30% - **Critical minerals (In, Y, Co, Ti) recovered** via ionic liquid extraction

End of Way-Finder Haptic Insoles Cost Analysis | CSMFAB 038 | June 2026